

ANTONIO RAPUANO

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SUMMARY

I'm an engineer driven to turn theory into practical systems. I enjoy full-stack robotics — working across the development pipeline, from mathematical modeling and control design to implementation and experimental validation. I'm committed to advancing autonomous systems through research and hands-on engineering.

EXPERIENCE

Research Engineer (Research Fellow) Aug 2025 - Apr 2026

Sapienza University of Rome (uniroma1.it)

Visiting period: University of Twente (utwente.nl) Jan - Mar 2026

Highlights: Design and implementation of novel aerial interaction control strategies • Development of tactile sensing models using machine learning • Experimental validation on a hexarotor platform

Autopilot Control Systems Engineer Oct 2024 - Jul 2025

Sapienza Flight Team (sasa-aerospace.it/flight-team)

Highlights: Design and implementation of path planning and hybrid control algorithms • Development of state estimation and filtering modules • Experimental testing on VTOL prototype

Race Timing Specialist and Team Lead Sep 2018 - Present

ICRON (icron.it)

Highlights: Coordination of race logistics and technical setup • Supervision of start/finish line operations • Management of ranking and award procedures

EDUCATION

Ph.D. in Automatic Control, Bioengineering and Operations Research Nov 2025 - Present

Sapienza University of Rome, Rome, Italy

M.Sc. in Control Engineering Sep 2022 - Jul 2025

Sapienza University of Rome, Rome, Italy

Final grade: 110/110 with honors

Thesis: Nonlinear predictive control of the continuum and hybrid dynamics of a suspended deformable cable for aerial pick and place (Advisor: Prof. Antonio Franchi)

B.Sc. in Electronic Engineering for Automation and Telecommunications Sep 2018 - Feb 2022

University of Sannio, Benevento, Italy

Final grade: 110/110 with honors

Thesis: Implementation on FPGA of an electronic circuit controlling a mechanical arm (Advisor: Prof. Marco Pisco)

PUBLICATIONS

A. Rapuano, Y. Shen, F. Califano, C. Gabellieri, and A. Franchi. "Nonlinear Predictive Control of the Continuum and Hybrid Dynamics of a Suspended Deformable Cable for Aerial Pick and Place." *IEEE ICRA*, 2026. arxiv.org/abs/2602.17199

SKILLS

Software and tools: Arduino • Blender • Fusion • Gazebo • LTSpice • Quartus Prime • ROS 2 • Simulink • Stateflow

Programming and formatting: C/C++ • Git • HTML/CSS • Java • LaTeX • MATLAB • Python • Verilog

Hardware: 3D printing • Circuit board soldering • Electronic measurement instruments

Personal: Attention to detail • Autonomy and initiative • Critical thinking • Curiosity and continuous learning • Problem-solving attitude • Project planning • Technical writing • Time management

LANGUAGES

Italian, native

English, fluent

Cambridge English C1 Advanced - Score 207 (Grade A, CEFR Level C2)

Jul 2022